

# AbdulHamid Mostafa Gouda

## Software Engineer

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## Summary

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Backend-focused full-stack engineer and former ICPC community leader. Graduated with Honors in Computer and Control Systems Engineering and completed ITI's 9-month full-stack program, building scalable REST APIs and multi-role web applications across Node.js, Django, and Laravel.

## Education

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### 9-Month Trainee, Open Source Application Development

Oct 2025 – Present

Information Technology Institute (ITI), Mansoura

- Completed training in Node.js, React, Angular, Vue.js, NestJS, Next.js, GraphQL, PHP, Python, and Java
- Built 5+ RESTful APIs using MongoDB and MySQL; applied Agile, design patterns, testing, and Docker

### B.Sc. in Computer and Control Systems Engineering

Sep 2020 – Jul 2025

Mansoura University

- Graduated with Excellent with Honors distinction
- Graduation Project: HealthMate🔗, a Health Tracking and AI Insights System
- Studied data structures, algorithms, operating systems, networking, and software engineering principles

## Experience

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### Programming Instructor, Part-Time, Futuristic Organization

May 2025 – Aug 2025

Mansoura, Egypt

- Taught programming to 20+ students aged 8 to 18, covering HTML, CSS, JavaScript, and Scratch
- Designed a project-based curriculum that improved student engagement and concept retention
- Simplified core programming concepts, enabling beginners to build their first web projects within 4 weeks

## Projects

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### Clinic Scheduling Management System🔗

Vue.js, Django REST Framework, JWT, MySQL

- Built a full-stack healthcare application supporting appointment booking for 3+ user roles (patient, doctor, admin)
- Implemented real-time physician availability tracking, reducing scheduling conflicts by design
- Designed 10+ RESTful API endpoints to manage scheduling logic and patient-doctor interactions

### Node.js Authentication System with Two-Factor Authentication🔗

Node.js, Express.js, MongoDB, Passport.js, JWT

- Architected a secure authentication system supporting JWT, OAuth 2.0, TOTP-based 2FA, and social login via Google and Facebook
- Reduced unauthorized access risk by implementing rate limiting, brute-force protection, and secure cookie handling
- Built email and SMS verification flows covering registration, password recovery, and account management

### E-Commerce Platform API🔗

Node.js, Express.js, MongoDB, Mongoose, Stripe, Nodemailer

- Developed a production-ready REST API handling user authentication, product management, and order processing
- Integrated Stripe payment gateway and Nodemailer for transactional emails, supporting end-to-end purchase flows
- Designed scalable Mongoose schemas and enforced clean architecture with modular route and controller separation

### Hotel Management System🔗

Laravel, Vue.js, MySQL, Tailwind CSS

- Delivered a full-stack hospitality platform covering room management, reservation workflows, and admin controls
- Achieved a fully responsive UI using Tailwind CSS and ShadCN, cutting UI development time significantly

## RAG-Powered Document Q&A Platform

*FastAPI, LangChain, OpenAI, MongoDB, React, Vite*

- Built a full-stack RAG pipeline for uploading PDF/DOCX/TXT documents and querying them via natural language, powered by LangChain orchestration and OpenAI embeddings
- Engineered FastAPI REST API with document ingestion and semantic chat endpoints, paired with a React + Vite frontend — developed entirely via **Vibe Coding**

## Skills

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**Programming Languages:** JavaScript, TypeScript, Python, PHP, C++, Java

**Backend:** Node.js, Express.js, NestJS, Django, Laravel, RESTful APIs, GraphQL

**Frontend:** HTML, CSS, Tailwind CSS, Bootstrap, Vue.js, React, Next.js, Angular

**Databases and ORMs:** MongoDB, MySQL, Redis, Prisma, Mongoose, Eloquent

**Tools:** Git, GitHub, Postman, Docker, Linux, Bash

**Concepts:** Data Structures and Algorithms, Object-Oriented Programming, Design Patterns, Agile

**AI and ML:** Prompt Engineering, Retrieval-Augmented Generation (RAG), AI API Integration

## Activities

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**Vice Leader, ICPC Mansoura Community** ([msc.acm.org](https://msc.acm.org))

*Aug 2023 – Oct 2024*

Mansoura, Egypt

- Organized 3 training levels across 3 faculties, reaching 700+ participants
- Led a team of 100+ members and directed Game of Coders 3.0, the largest competitive programming event in the Delta region
- Coached and prepared ECPC-qualifying teams, contributing to improved contest performance

**PR and FR Head, ICPC Mansoura Community**

*Feb 2023 – Aug 2023*

Mansoura, Egypt

- Secured 5+ sponsorships and partnerships by preparing professional proposals and managing external relations
- Coordinated logistics for 3+ events including speaker scheduling and permit acquisition

**FR Supervisor, CAT Reloaded**

*Aug 2023 – Sep 2024*

Mansoura, Egypt

- Mentored 10+ team members on responsibilities and workflows; delivered bi-weekly performance reports to management

## Courses and Certifications

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- Node.js, Express, and MongoDB Bootcamp, Udemy (Jonas Schmedtmann) – Built a full tour booking API
- Web Development using PHP (150 hours), Information Technology Institute (ITI)
- Front-End Web Fundamentals (60 hours), Information Technology Institute (ITI)
- Building Web Applications using PHP and MySQL, MaharaTech
- Competitive Programming Training Level 2, Coach Academy EG
- Ideation BootCamp, Creative Innovation Hub – Design Thinking, Business Model Canvas, Pitch Deck
- Entrepreneurship, TIEC
- Python Django – The Practical Guide
- Docker from Zero to Hero
- Core Java Programming

## Competitive Programming

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- Participated in ECPC Qualifications twice 2022 and 2023
- Solved 500+ problems on Codeforces and Vjudge; also active on LeetCode
- ICPC Global Profile

## Languages

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**Arabic:** Native    **English:** Proficient